“Observation of species in the wild convinces me that the existence and persistence of species is vitally bound up with environment.”

Joseph Grinnell

Niches and competition

British-born zoologist G. Evelyn Hutchinson, working at Yale University from the 1950s to the 1970s, examined all the physical, chemical, and geological processes at work in ecosystems and proposed that any organism’s role in its niche includes how it feeds, reproduces, finds shelter, and interacts with other organisms and with its environment. For example, each species of trout—and other fish—has its own range of water salinity, acidity, and temperature that it can tolerate, as well as a range of prey and river- or lake-bed conditions. This makes some better competitors than others, depending on the conditions of the habitat in which they live. Seen as the father of modern ecology, Hutchinson inspired other scientists to explore how competing animals use their environment in different ways.

An animal or plant’s niche width comprises the whole range of factors it requires to allow it to thrive. Brown rats, raccoons, and starlings are examples of animals with a broad niche width in that they are able to survive in a wide variety of conditions. Such species are called generalists. Other animals have narrow requirements. For example, koalas depend almost entirely on eucalyptus leaves, and Hyacinth Macaws in the Pantanal region of Brazil eat virtually nothing but the hard fruits of two species of palm trees—these are specialists.

Animals rarely occupy the whole of their niche width, owing to competition between species. Part of the habitat requirement of North American bluebirds is dead trees with old woodpecker holes in which they lay their eggs and raise their young. Although suitable holes are common in many forests, bluebirds cannot occupy all these holes because they are often